Self-Sustaining Tritium Regeneration Loop in Proton–Electron Capture Chain Reactions: A Closed Fuel Cycle for Fusion Energy

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Abstract

We demonstrate theoretically that the proton–electron capture chain mechanism (PECCNAL) produces a self-sustaining tritium regeneration loop requiring no external tritium supply. Tritium consumed in deuterium–tritium fusion (Step 3) is continuously regenerated via two independent pathways: neutron–deuterium capture (Step 2) and Lithium-6 neutron capture (Step 4). The net result is a closed fuel cycle in which only deuterium and Lithium-6 are consumed, while tritium acts as a catalytic intermediate. This resolves the central fuel supply problem of D–T fusion energy.

Keywords: tritium breeding, closed fuel cycle, self-sustaining loop, D–T fusion, neutron capture, lithium-6

1 Introduction

Tritium (^{3}H) is an essential fuel for D–T fusion but is radioactive (half-life 12.3 years) and not naturally abundant. Current fusion programs rely on external tritium supply or blanket breeding, neither of which achieves true fuel self-sufficiency.

In this paper, we show that the PECCNAL mechanism [?, ?] naturally produces a closed tritium loop, where tritium is regenerated at the same rate it is consumed.

2 The Tritium Loop

The complete cycle is:

$$p + e^- \to n + \nu_e \quad - 1.293\,\mathrm{MeV} \tag{S1}$$
$$n + {}^2\mathrm{H} \to {}^3\mathrm{H} + p \quad + 2.220\,\mathrm{MeV} \tag{S2}$$
$${}^3\mathrm{H} + {}^2\mathrm{H} \to {}^4\mathrm{He} + n \quad + 17.600\,\mathrm{MeV} \tag{S3}$$
$$n + {}^6\mathrm{Li} \to {}^4\mathrm{He} + {}^3\mathrm{H} \quad + 4.780\,\mathrm{MeV} \tag{S4}$$

Table 1: Tritium production and consumption per cycle

Step	**Tritium**	**Net**
S2: $n + {}^2\mathrm{H} \rightarrow {}^3\mathrm{H}$	+1 produced	
S3: ${}^3\mathrm{H} + {}^2\mathrm{H} \rightarrow {}^4\mathrm{He}$	-1 consumed	
S4: $n + {}^6\mathrm{Li} \rightarrow {}^3\mathrm{H}$	+1 produced	
Net per cycle		**+1 surplus**

2.1 Tritium Balance

The system produces *more* tritium than it consumes, achieving a tritium breeding ratio (TBR) > 1:

$$\mathrm{TBR} = \frac{T_{\text{produced}}}{T_{\text{consumed}}} = \frac{T_{S2} + T_{S4}}{T_{S3}} = \frac{1+1}{1} = 2 \tag{1}$$

A TBR of 2 means the system doubles its tritium inventory per cycle, enabling exponential scale-up without external supply.

3 Energy Balance of the Closed Loop

$$E_{\text{net}} = -1.293 + 2.220 + 17.600 + 4.780 = \boxed{+23.307\,\text{MeV per cycle}} \tag{2}$$

4 Fuel Consumption

Per complete cycle, the system consumes:

- 2 deuterium nuclei (${}^2\mathrm{H}$): one in S2, one in S3
- 1 Lithium-6 nucleus in S4
- 1 electron in S1 (replenished by ionization)

Tritium is **not consumed** on net — it is a catalytic intermediate regenerated each cycle.

5 Comparison with ITER Blanket Breeding

6 Conclusion

The PECCNAL mechanism achieves a self-sustaining tritium regeneration loop with TBR = 2, requiring only deuterium and Lithium-6 as consumed fuels. This eliminates the tritium supply bottleneck that currently limits D–T fusion development.

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Table 2: Tritium breeding comparison

Feature	**ITER Blanket**	**PECCNAL Loop**
TBR target	> 1.05	2.0 (theoretical)
External T supply needed	Yes (startup)	No
T regeneration pathway	Single (Li blanket)	Dual (S2 + S4)
Coulomb barrier	N/A	None
Scale-up	Linear	Exponential

References

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